Commercialization Plan

AccuGyd: Accelerated Guidance for Accurate MRI-Guided Neurosurgery

Overview: AccuGyd is an innovative surgical tool that facilitates precise guidance of surgical devices into the brain. For many types of therapies and procedures, accurate placement is key to a good outcome, but speed and simplicity determine whether new groundbreaking therapies are ultimately adopted. AccuGyd was specifically designed to address shortcomings of current competing devices: it is compatible with MRI or CT image guidance; it is much faster to deploy and align; and it is significantly more compact, while maintaining similar or better accuracy. This SBIR proposal aims to further develop and validate the AccuGyd device with the aim of submitting it for FDA 510(k) clearance by project's end.

AccuGyd was inspired by frustrations that Dr. Azam Ahmed, a neurosurgeon and co-founder of ImgGyd LLC, had with an existing product, the ClearPoint™ guidance device. The shortcomings that neurosurgeons find with ClearPoint™, the acknowledged market leader in this space, are described in detail in Section A. Our improved design is described in Section A.3, with a more detailed description in Section 3 of the Research Plan. We view having a competitor in an existing market as a strong positive, for several reasons: 1) Instead of inventing a conceptually new product, we rely on the competing product's proven paradigm while proposing a solution addressing the pain points of existing customers; 2) We do not have to create a new market; 3) Estimates of market size and cost structures are readily obtained; and 4) There is an established precedent for FDA 510(k) clearance. **We are placing a superior product into an existing ecosystem, rather than inventing an entirely new market.**

In the medical device market, obtaining FDA clearance is a difficult but crucial hurdle to overcome. Once obtained, it can greatly shrink the competitive field. Our early plan to market, explained in detail below, has three phases: 1) product development and pre-clinical testing; 2) FDA 510(k) clearance; and 3) deployment with select luminary sites. This approach allows AccuGyd to obtain important **early traction, valuable experience, and early sales** in pre-clinical (non-human) research, an infeasible approach with many medical devices. Second, the image-guided neurosurgery field is relatively small, so placing AccuGyd at as few as five luminary sites, ideally teaching institutions, can be a powerful force multiplier. Our Letters of Support indicate interest from pre-clinical researchers as well as numerous neurosurgeons who indicate they want to use AccuGyd when it becomes available.

In 2023, ImgGyd participated in an entrepreneurial bootcamp: the Wisconsin SBIR Advance program, sponsored by the Wisconsin Economic Development Corporation (WEDC). The program is based on the popular Stanford small business program, similar to the I-Corps program offered by the NIH. Customer discovery was the main focus; we conducted over 50 structured interviews of neurosurgeons and other stakeholders in image-guided neurosurgery. These interviews have been invaluable in shaping our plans for device development and marketing approach. While we were surprised by the diversity of approaches in neural device placement, we feel that we have a good initial grasp of the market and its needs. We also feel confident that **AccuGyd addresses a real need, and customers will choose our device over current competitors.**

The current market centers around placement of deep brain stimulator (DBS) devices to manage **Parkinson's disease** (PD), and **ablation of epileptic foci**. Additional applications include the placement of electrical devices to manage epilepsy and treating neurological cancers. **Stroke therapies** could emerge as a focus, depending on the results of the REpeated ASSEssment of SurvivorS in Intracerebral Hemorrhage Study (REASSESS), which is poised to examine stroke clot removal via minimally invasive surgery (MIS). A game-changing potential application is **gene therapy**, where therapies must generally cross the blood-brain barrier to enter the brain parenchyma. The NIH and private industry are developing many gene therapy pharmaceuticals, in a field with potential to become a very large and lucrative market (discussed below). Gene therapy's huge promise is a likely driver of recent major investments in this area, but many of these therapies are hindered by challenges in treating large brain volumes to achieve efficacy. In our market analysis below, we demonstrate a current market that is realistic and profitable, and a growing market for gene therapy and stroke therapies that collectively could provide a much larger market for AccuGyd. We are excited to be entering the market now, since we feel that AccuGyd can meet customers' existing needs while getting in on the ground floor of exciting new therapies.

The pricing and sales structure for AccuGyd is based on the analysis of the market for the main competitor device, ClearPoint™. Our research, based on publicly available data, indicates a per-procedure charge of $5000-$6000. We plan to charge at the low end of this range ($4000). A Serviceable Available Market (SAM) of $300 million annual revenue is justified below. Capturing a small fraction of this would be a major commercial success.

Objectives

A. Value of the SBIR Project, Expected Outcomes, and Impacts.

A.1 Introduction. The AccuGyd technology was developed in response to demands from neurosurgeons for a faster and less complex alternative to ClearPoint(1-5). This sentiment was summarized by a comment from one neurosurgeon regarding the ClearPoint device he currently uses for MRI-guided procedures: "When we first got ClearPoint, I loved it. Now I hate it." This is another way of saying that ClearPoint solves a problem and is the current best tool for the job, but with familiarity comes a realization of its weaknesses. Though ClearPoint was groundbreaking and eagerly adopted when it was introduced, subsequent technical development has been limited. The main weakness is the unpredictable number of iterations needed to set the ClearPoint device to the desired target trajectory. **AccuGyd was designed to require only a single non-iterated step** to align the device trajectory with the target. The neurosurgeons who have seen AccuGyd are enthusiastic about it, citing its small footprint, ease of use, and potential to significantly reduce procedure times.

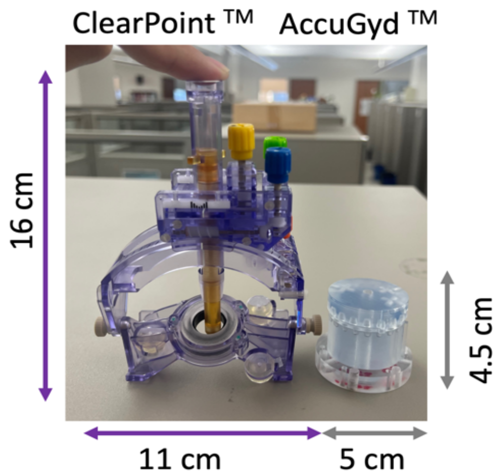


Figure 1: Comparison of ClearPoint and AccuGyd devices.

AccuGyd has 1/8 the volumetric footprint. The indexing system eliminates moving parts, contributing to excellent rigidity. The portion of both designs that is responsible for stabilizing the inserted drill or catheter is quite similar. However, the moving armatures in ClearPoint's design create a large additional structure.

A.2. Background. As surgeons became familiar with ClearPoint™, the current market leader in MRI-guided neurosurgery, they identified several weaknesses. AccuGyd was developed to address these weaknesses. Both devices are shown in Fig. 1. AccuGyd's workflow is discussed in the Research Plan (Fig. 1, Section 3.2); a brief discussion of ClearPoint's workflow follows.

The workflow for the ClearPoint™ tower is illustrated in Fig. 2. The location to place the tower ("entry planning") is determined. A 13mm craniotomy access hole is created for most procedures. The base is then secured to the skull with three skull screws and three standoff screws. The surgical and radiology teams then commence an iterative series of steps: 1) MRI imaging data informs the ClearPoint laptop of the current trajectory of the tower; 2) ClearPoint software calculates and suggests manipulations of its four colored knobs to align the tower with the brain target; 3) the surgeon manipulates the knobs; 4) ClearPoint software directs the radiology team to acquire images in the estimated vicinity of the tower; and 5) return to step 1. For most surgical teams who use the system only a few times per month, the process of installation and alignment takes 3-6 iterations, or typically about two hours per tower installation.

The many moving parts inherent to the design necessitate multiple iterations to refine the trajectory alignment. These parts suffer from hysteresis, or an inability to return predictably to a precise starting point. Thus, the surgeon moves the ClearPoint trajectory back and forth across the desired trajectory until "close enough" is achieved.

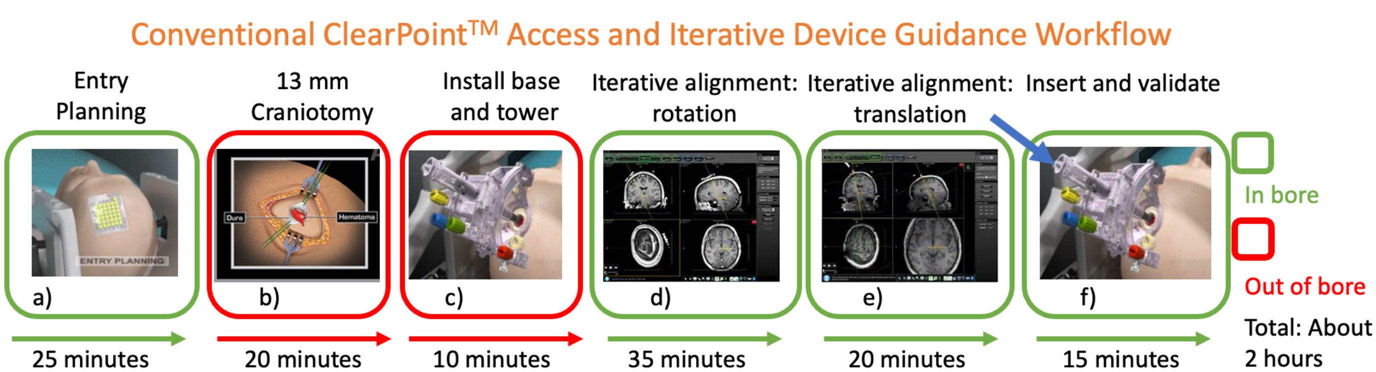


Figure 2: The predominant MRI guidance system, ClearPoint, requires ~2 hours to install and align the trajectory in order to insert a device through a 13mm burr hole. a) MRI visible planning grid; b) 13mm burr hole craniotomy; c) Installation of the base and tower; d,e,f) Guidance is performed by iterations of scanning, computation, and manipulation of the guide by rotation and translation before finally inserting the catheter or electrode. Every time a knob is turned, imaging is required.

A.3. Gap Analysis of the ClearPoint™ System. Several of the following gaps (1-5) summarize shortcomings identified by early adopter neurosurgeons of the ClearPoint platform. These gaps are readily addressed by the innovations in our AccuGyd device. Gaps 6-8, and to some extent gap 4, are related to issues with the overall workflow and environment for MIS. By simplifying the workflow and significantly shortening the procedure time with AccuGyd's approach, all of these gaps can also be addressed.

Gap 1: Poor Rigidity. ClearPoint™ lacks the mechanical rigidity to maintain alignment while guiding a small drill bit or device through the skull. ClearPoint claims 2mm drill holes can be achieved, but its imprecise gears and the tall, non-rigid tower fail to stabilize the hand drill. This leads to an aberrant pathway through the skull that in turn defines an inaccurate trajectory. Our LOS from Dr. Lake summarizes why larger burr holes are thus required.

Gap 2: Large Skull Opening. Lacking a precisely aligned 2mm drill hole, surgeons instead resort to a large 13mm hole so the skull won't interfere with the catheter. Clinical benefits of small holes include better infection control, faster wound healing, and reduced need to periodically remap a slumping brain due to CSF drainage.

Gap 3: Infinite Adjustment. ClearPoint's design theoretically permits an infinite number of trajectories within its range of motion. However, hysteresis in the imprecise gears leads to a tedious iterative alignment process, with multiple rounds of scanning, trajectory updates, and manipulating the device. As one of our neurosurgeons commented, "Infinite adjustment possibility means infinite adjustment time!"

Gap 4: Long, Unpredictable Procedure Times. In standard OR procedures such as placing surveillance electrodes, neurosurgeons insert electrodes through 2-3mm drill holes within 10 minutes. Neurosurgeons use intra-operative MRI, often with ClearPoint™, when they need sub-mm accuracy. ClearPoint users typically need three to six rounds of up to 10 minutes each for alignment, with the ClearPoint alignment portion taking 2 hours, and the entire catheter insertion process taking 3-4 hours hours. Our interviews with anesthesiologists showed how increased surgical time 1) increases interruptions in the surgical process to check the patient more frequently; 2) significantly increases risk of post-surgical cognitive impairment and imposes other systemic stresses, especially in older patients; 3) linearly increases their billed efforts; and 4) hinders the hospital to staff other surgical procedures and predict operating room availability.

Gap 5: Too large for Pediatric or Non-Human Primate Studies. The footprint of ClearPoint is too large to be installed on smaller heads, including infants, smaller children, and non-human primates such as macaque monkeys. Forthcoming gene therapy treatments for childhood diseases will require a solution sized appropriately for these patients.

Gap 6: Intraoperative Guidance with MRI. Surgeons typically use MRI for pre-surgical planning and post-surgery validation. They would prefer to use MRI during surgery, but many surgeons' experience has been poor: The neurosurgeons on our team have emphasized, sometimes at great length, their frustration at the long setup time needed for current MIS guides. While surgeons appreciate the increased accuracy afforded by MRI guidance, they recognize that adding 60 minutes to an already complex procedure may not improve outcomes.

Gap 7: Dual Use in CT and MRI not Supported. Current image guides are designed for either MRI or OR environments. The OR devices tend to be more advanced, but generally cannot enter the MRI scanner, while MRI guidance devices are limited because they cannot include ferrous components. No current MRI guidance devices would be the first choice for a procedure performed in a non-MRI environment.

Gap 8: Limited Customer Base. Beyond major academic centers, few neurosurgeons use ClearPoint's complex workflow. A large surgical team (6-14 personnel) is needed with support from radiology, medical physics, anesthesiology, and neurosurgery; such teams are only available at major research hospitals.

A.4 Solution. AccuGyd was initially designed in response to Gap 4, the long and unpredictable procedure times that our surgeon, Dr. Ahmed, found so frustrating. We found two major sources: Gap 1, the poor rigidity of the competitor, and Gap 3, difficulty aligning the correct trajectory due to the inherent hysteresis in the complex moving parts. With the benefit of hindsight, AccuGyd was designed expressly to address and rectify all eight of the Gaps listed above. AccuGyd is designed to be better, faster, less expensive, easier to use, and to fit naturally into established neurosurgical workflows.

A.5 Innovation. The AccuGyd device, while elegantly simple, contains several novel, innovative features. The patented approach uses a finite indexed mechanism which requires no adjustment. AccuGyd uses a guidance cassette that is installed at a specific index location into the registered base. (The mathematical framework for this innovation is provided in Section 3 of the Research Plan.) **Changing the trajectory on the fly is simple**, especially if the base is not moved: Specify a new target, calculate a new trajectory, and adjust the new AccuGyd settings, all in under 5 minutes. The trajectory alignment is rapid, easy, and precise.

The indexed design enables small 2mm drill holes, which to date have eluded most neurosurgeons. Larger or smaller holes can easily be accommodated. The small footprint is compatible with most CT scanners, MRI scanners, and MRI head coils. The small size is transformative for pediatric gene therapy trials (see LOS from Thomas Miller, Bayer, and Stanford Pediatric Radiology Chief Shreyas Vasanawala). Our proposal also features a novel head phantom design, which may prove useful as a product in its own right for training purposes.

A.6 Expected Outcomes. The main outcome of this Phase II project will be a comprehensive image guidance device that is ready to enter application for FDA 510(k) clearance. While the basic principle of an indexed alignment system has been demonstrated in Phase I, the activities presented in the Research Plan are needed to create a commercial product that has been adequately tested and validated, with excellent supporting hardware and software. In our discussions with neurosurgeons, it was immediately apparent that AccuGyd will never be used without a complete hardware kit and excellent display capabilities. We will work with a software developer, Sanctuary Software Solutions (see quotation letter), with extensive experience in developing medical-grade software for FDA-approved medical devices. We will also retain an established Medtech CRO firm, NAMSA, to develop our regulatory strategy and ensure that the software and hardware developed here is compliant with relevant FDA guidance and regulations. **At the end of this 30-month Phase II project, we will be ready to submit AccuGyd for FDA 510(k) clearance.**

A.7 Impacts. The AccuGyd guidance device is likely to have three major impacts. First, by reducing the complexity and duration of image-guided neurosurgeries, it can **lower barriers to performing image-guided surgeries.** This will result in lower costs, better facility utilization, and improved patient outputs (see Garry Gold LOS, Chair of Stanford Radiology). Second, few neurosurgeons and facilities can now perform complex image-guided minimal invasive neurosurgeries; demand for these procedures is likely to grow, but current procedural complexity and the limited number of facilities will create a bottleneck for the total number of procedures nationwide. By reducing procedure time and complexity, current facilities can increase throughput, and **additional facilities are likely to consider adding such procedures**. Third, by providing a much smaller device, **image-guided procedures on pediatric populations are facilitated**. This development will support delivery of gene therapies for neurodegenerative orphan diseases, where very early intervention is critical. The many ongoing and planned NIH-sponsored clinical trials of gene therapies (detailed below) could see major benefit from a faster procedure with a much smaller footprint.

A.8 Integration with Overall Business Plan. AccuGyd is our main product, and this SBIR funding will enable us to ready AccuGyd for the FDA clearance process. We envision AccuGyd as the centerpiece of a sophisticated image guidance platform with opportunities for sales of related products, including improved therapy catheters, therapy monitoring software, novel skull drills, and improved sterile drapery. The drapery product, “Saf-T-Drape”, is likely to become commercially viable before AccuGyd, due to reduced regulatory burden. AccuGyd customers are likely to purchase our drapery system, which will be an excellent marketing angle for both products.

B. Company Overview

B.1 Goals. **Our strategic goal is to create a portfolio of hardware and software technologies that will provide a strong basis for a comprehensive image-guided platform**, while also providing a strong IP position with products that can be brought to market in a well-defined timeframe. ImgGyd focuses on creating products that can improve the workflow and user experience for image-guided neurosurgery. This is a natural fit for our founders’ research and professional interests, allowing us to rely on our deep expertise in this specialized area.

B.2 History. ImgGyd LLC is a small medical device startup company in Madison, Wisconsin. The four founders have worked together on many projects over the last 10-20 years, with a primary focus on medical imaging and image-guided neurosurgery. Our previous company, Insert MRI, Inc., obtained several SBIR grants focused on developing a real-time MRI NeuroTherapeutic Guidance platform. This platform was a technical success, but the path to independent commercial success was hampered by too many interdependencies with pharmaceutical programs, therapeutic clinical trials, scanner vendor hardware platforms, and the adoption of specific neurosurgical procedures. Following the successful sale of Insert MRI, Inc., we focused our efforts on the hardware and software components that are currently impediments to more widespread adoption of image-guided neurosurgery. We formed a new company, ImgGyd LLC, with a streamlined corporate structure and a tighter focus on smaller pieces of technology that we can directly control, such as AccuGyd.

ImgGyd LLC benefits from the experience of the previous company, including a history of successful SBIR and other NIH grants, good working relationships with several leading companies in the image guidance space, and experience in building and running a company. The experience of selling our previous company was invaluable: we now have a **much tighter focus on the short- and long-term items that are needed to create a product, move it to market, and create value for the company**. We added a key new team member, ImgGyd founder

Dr. Azam Ahmed, a neurosurgeon at UW Hospital with whom we have worked closely on several projects. Drs. Alexander and Block have several current and recent NIH-sponsored grants relevant to the interests of ImgGyd, and we are actively working to move these academic ideas into the commercial product space.

B.3 Team.

B.3.1. The ImgGyd Team consists of 4 founders:

- Azam Ahmed, MD, is a neurosurgeon at UW-Madison Hospitals. He participates in several large research studies, serving as Regional Director of the NIH StrokeNet and co-investigator in the Phase 2 and Phase 3 portions of the NIH Minimally Invasive Surgery Plus Alteplase for Intracerebral Hemorrhage Evacuation (MISTIE) trials, and the related REASSESS trial. Dr. Ahmed is ImgGyd's surgical insider, who can translate his experiences and those of his colleagues into product ideas to meet current gaps in surgical hardware.
- Andrew Alexander, PhD, is a Professor of Medical Physics with a research career focused on advanced MRI applications. He was one of the developers of an early MRI guidance system for pre-clinical research. He is Senior Fellow of the International Society of Magnetic Resonance in Medicine (ISMRM) and recognized for his expertise in diffusion tensor brain imaging, a key expertise needed for surgical planning in pediatric gene therapy trial development.
- Walter Block, PhD, is a Professor of Medical Physics and Biomedical Engineering. He is an expert in MRI imaging and is interested in translating complex MRI techniques into usable clinical procedures. Dr. Block worked for General Electric Medical Systems (GEMS) for the first part of his career, acquiring insight into the business side of medical imaging. His group guided seminal gene therapy experiments in bio-psychiatry(6) and stem cell neuro-integration as featured in *Nature Medicine*(7). A Senior Fellow in the ISMRM, he is an inventor on 15 imaging patents.
- Terrence Oakes, PhD, is a medical physicist who has focused on quantitative and semi-quantitative imaging methods in MRI and PET. Dr. Oakes has led several large research programs in academia, government, and private industry. Previous to joining ImgGyd, he was the CEO of Insert MRI Inc., and led a successful sale of the company. He is now the president of ImgGyd and participates actively in research as well as in administration and business development.

Drs. Alexander and Block founded an earlier company, Insert MRI, Inc., in 2011. They hired Dr. Oakes to manage and run that company, which underwent a successful sale. Based on their good working experience, Alexander, Block and Oakes launched another company, ImgGyd LLC, adding Ahmed as a fourth founder.

UW-Madison was the second clinical ClearPoint™ site after UCSF where the technology was created. The co-founders of ImgGyd, in our roles at UW-Madison, were present for discussions outlining the various training protocols, the certification of UW-Madison as a ClearPoint clinical site, and the first clinical procedures using ClearPoint for DBS insertions. ClearPoint is independent of the scanner manufacturer software platforms, and we intend to follow the same route. We learned which procedural components needed delineation by ClearPoint and which parts the surgical team had to initiate, while remaining within the boundaries of certification from ClearPoint. Dr. Ahmed was one of the first surgeons to use the ClearPoint tower in Visualase™ focal epilepsy procedures after Medtronic acquired Visualase.

Working with ClearPoint, we participated in developing surgical teams and responsibilities, surgical and imaging protocols, and fallback strategies. Our feedback was used in developing the first revision of the ClearPoint instruction manuals. We became very familiar with the background, teaching style, scheduling philosophy, and problem resolution strategies from ClearPoint's very capable Head Application Specialist who personally handled all matters for getting the first 5 customers up and running. In short, there are common aspects to any introduction of a new surgical product, but the introduction of devices for MRI- and CT-guided procedures has distinctive elements which the ImgGyd co-founders were able to observe, experience, and contribute firsthand.

Our team has worked together in the neurosurgery and MRI guidance space for over 10 years, with individual experience spanning decades. We have expertise in targeted convection-enhanced delivery (CED) infusions, minimizing backflow along the catheter exterior (8), minimizing infusate loss along white matter tracts, and sophisticated MRI acquisition and display techniques. With the Wisconsin Alumni Research Foundation (WARF, the University's IP arm), we developed, patented, and licensed technology for rapid trajectory planning and device guidance for a generalized brain guide(9, 10). Our previous company developed a novel platform for real-time control of the MRI scanner to guide and monitor delivery of neurotherapeutics into the brain. This platform guided CED delivery of stem cells(7, 11), gene therapies(6, 12), and other drugs(13) into precise brain locations of >100 nonhuman primates. **The objective of our new company is to provide technologies that will *simplify, accelerate and minimize the invasiveness* of MIS neurosurgical procedures**.

B.3.2. Partners, Collaborators, Consultants. Since ImgGyd is not the first business we have started, we already have a strong team for the administration aspects, including a local business bank, a legal firm that focuses on small business and entrepreneurs, and an excellent accounting firm. There is strong support for entrepreneurs and technology-focused startups at UW-Madison as well as in the greater Madison area, including active mentor programs, bootcamps for product development and customer discovery, business-focused startup grants, and excellent access to research and office facilities.

We have worked with several consultants on business development projects, both in our previous and current companies. We obtained funding through the Wisconsin Economic Development Corporation to work on brand development and to develop a comprehensive financial plan with our previous company. Each of these activities were multi-month projects, and the experiences are readily applicable to our current company, ImgGyd.

Customer Discovery Interviews. ImgGyd obtained funding from the Wisconsin SBIR Advance program through WEDC in May 2023 to participate in a 3-month entrepreneurial bootcamp. The program is based on the popular Stanford small business program, similar to the I-Corps program offered by the NIH. We obtained assistance in researching our competitive landscape and developing presentations. However, the main focus was on customer discovery, and **we conducted over 50 structured interviews of neurosurgeons and other stakeholders in image-guided neurosurgery.** We now have a good grasp on the market and on customer needs, having interviewed users of AccuGyd (neurosurgeons), referrers for neurosurgery procedures (neurologists and oncologists), procedure participants (anesthesiologists, radiologists, MRI and CT experts, radiology techs), adjacent or competing providers (interventional radiologists), procedure gatekeepers (hospital billing personnel, hospital purchasing committee members), and regulatory experts. We also interviewed manufacturers, pre-clinical (animal model) researchers, and medical device executives. Several of the neurosurgeons who were interviewed became excited enough about AccuGyd to write a Letter of Support. We found this process to be extremely useful and informative, providing invaluable information about our customers and the market.

Partners. ImgGyd has a longstanding collaboration with Therataxis, a Baltimore-based company that has created iPlanFlow™, an industry-leading flow modeling program to plan and predict Convection-Enhanced Diffusion (CED) administration of neural therapeutics. Raghavan first licensed iPlanFlow™(14)to BrainLab for localized chemotherapeutic treatment of resected brain cancer margins and in recurrent brain tumors(15-17). Therataxis President Raghu Raghavan and ImgGyd co-founder Walter Block are empaneled on the FDA-supported Critical Path Consortium for Lysosomal Storage Diseases, which is working to streamline regulation and speed assessment of therapies for devastating neurodegenerative diseases like Batten Disease, MPS III, Gaucher's disease, and Niemann-Pick Disease. Therataxis is working on a next-generation iPlanFlow to predict infusion distributions in toddler brains. ImgGyd plans to incorporate this technology into our platform in future.

Consultants. Advice received from our legal team, our accountants, and our business mentors all point to minimizing the number of employees, instead hiring consultants or outside expertise as needed. We have developed reliable consultants and business relationships that we have used, and will continue to use, for CAD design and modelling, 3D printing, engineering and material design advice, phantom construction, software development, editorial services, and promotional materials. To date we have not felt the need to internalize any of these services. We were fortunate to engage Professor Michael Zinn(18) of the UW-Madison Department of Mechanical Engineering to assist us on a rigid design for AccuGyd. Prior to joining UW-Madison, Zinn was an engineer at Hansen Medical in Mountain View, CA, developing many of the patented ideas that were cross-licensed with Intuitive Medical to form the daVinci surgical robot.

B.3.3. Business Mentors. We rely on Joseph Boucher, Esq., of Neider & Boucher, a law firm specializing in guiding startup firms to exits for over 30 years in Madison. Boucher mentored us on legal and business aspects of our former company, a role he continues for our current company. **We are fortunate to be located in a region with significant success in imaging and neuromodulation startups**, including Rock Mackie's Pinnacle Imaging (now Philips) and Tomotherapy (now AccuRay)(19); Neuraworx (former UW BME Chair Justin Williams and Kip Ludwig), and Charles Mistretta's seminal development of digital subtraction angiography technology(21) and time-resolved MR angiography(22). These spinoff inventors are all members of our departments at UW-Madison, and they generously allow us to discuss our scientific and business needs with them regularly.

B.3.4. Future Hires. For the duration of this Phase II project, we will add one new employee, Thomas Lilieholm(23, 24), who participated in the Phase I AccuGyd project. Lilieholm is expecting to defend his PhD in July 2024, and is well poised to accelerate the technical development of AccuGyd. Most other Phase II efforts will rely on external consultants, though we will add several employees when we move closer to achieving FDA 510(k) clearance and there is a clear path to selling a commercial product. Initial hires will include a salesperson and an administrative assistant, followed by a VP-level Operations Manager. We will onboard sales and marketing personnel for anticipated growth into selected regions and markets. We plan to hire a marketing firm rather than internal employees, which will provide a much larger array of capabilities and expertise.

B.4 Regulatory Experience. We have a modest amount of experience in regulatory affairs. We worked with James Rogers of Coastal Consulting during our time with Insert MRI. Rogers successfully led the effort for HeartVista's FDA clearance of its real-time cardiac MRI platform(25), and we became familiar with the similar regulatory pathway needed for our product. Since AccuGyd does not depend on an MRI vendor platform or modifications to any MRI manufacturer platform, we are able to follow a much simpler path to FDA clearance than in our past experience. Dr. Oakes attended an "FDA Regulatory Education for Industry (REdI) Conference" for backgrounding on regulation of medical devices. Obtaining FDA clearance for medical devices requires highly specialized regulatory expertise. We intend to rely on NAMSA, an established Medtech CRO firm, for guidance in this area (see LOS), with an application for support from TABA funding to develop a comprehensive regulatory strategy with NAMSA. We anticipate that NAMSA will be positioned to guide us through the subsequent FDA 510(k) application process.

B.5 Vision and Future developments. ImgGyd is developing a number of products related to image-guided neurosurgery. **We envision AccuGyd at the center of a comprehensive guidance platform focused on providing a rapid, accurate and predictable workflow**, with a high level of accuracy and minimal personnel involvement. Other products under development include a novel drapery for MRI and CT scanners (patent pending); an advanced catheter design to limit backflow of injected therapeutics; a novel MRI-compatible drill; an MRI coil design optimized for image-guided neurosurgery; the MRI/CT phantom presented in this proposal, and adding Therataxis' intraparenchymal brain infusion drug distribution prediction software into a software product for guiding gene therapy trials.

Experience from our previous company taught us that it is not sufficient merely to have strong technical solutions and then hope others recognize them. A company must actively develop and promote its products in a competitive marketplace, taking active, difficult, and expensive steps to grow the company. We believe that ImgGyd's products can make a positive contribution to patient outcomes, and we are excited to **move our research endeavors into a commercial space where they can become available to clinicians and patients**.

C. Market, Customer, and Competition Analysis

C.1. Customers. The end customer for any medical procedure is the patient. However, for highly specialized surgical procedures, the patient may be consulted on broad choices but generally has little choice about specific pieces of equipment. The ultimate goal is to best meet patient needs by achieving the best outcome, so their care providers – the neurosurgeons – make the choices about the best equipment for the task at hand.

The customers for AccuGyd with actual buying power are neurosurgeons, the end users who can drive buying decisions related to their procedures. There are only about 3,500 neurosurgeons in the USA, with the majority performing procedures on the neck and spine. About 120-150 functional neurosurgeons in the USA regularly implant neurological hardware into the brain in complex surgeries, making for a compact and easily identifiable initial pool of customers. Most of these functional neurosurgeons are associated with advanced research-associated hospitals with abundant resources.

Surgeons, like most other professions, like to work with the equipment upon which they have trained. Accordingly, we plan to target several leading training centers for functional neurosurgeons as early sites for AccuGyd. However, surgeons are also generally open-minded about adopting new technology if the benefit is favorable. Based on our interviews with neurosurgeons and the excitement they showed about AccuGyd compared to similar alternatives, we believe the benefits that AccuGyd provide are more than enough to ensure that it will be positively received, since it will easily fit into and simplify existing surgical workflows.

Our market analysis also addresses an additional circle of influencers and stakeholders beyond the neurosurgeon customer base, discussed below.

Neurologists and oncologists are important parts of the patient referral process. They refer most patients for MIS neurosurgeries, and in some cases play a large role in deciding the treatment course, including choices about specific surgical techniques and equipment. Anesthesiologists are uniformly in favor of faster procedures, which reduce patient risk. MRI and CT technologists and physicists play a role in image acquisition and defining imaging protocols; this group favors shorter, more predictable procedures with well-defined imaging protocols.

Drilling through the skull is the most intense portion of a MIS procedure and requires increased anesthesia during that interval. Anesthesiologists have told us that there will be substantially less pain stress induced by the small 2-4mm drill holes made possible with ImgGyd, compared to the 13mm burr holes commonly required by

ClearPoint, and thus significantly lower anesthesia requirements. **Anesthesiologists are uniformly in favor of approaches and technology that reduce patient anesthesia dose and time**.

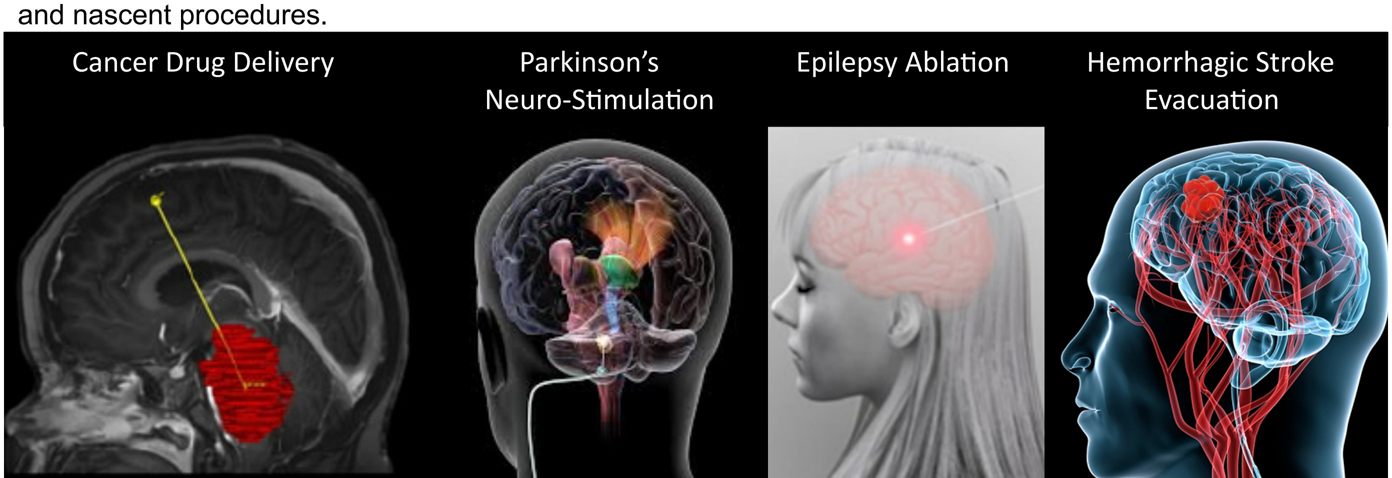
Providers of neuromodulation devices, such as Medtronic, indirectly affect the market as their application specialists often steer neurosurgeons in one direction or another regarding installation approaches. Discussions with the neuromodulation leadership team at Medtronic indicate that achieving a 2-hour or shorter procedure time for installing a DBS lead is likely an inflection point that would lead Medtronic to push for greater utilization of MRI in neuromodulation.

Hospital administrators, billing personnel, and radiology department heads seek cost savings achieved through lower equipment prices, but especially through reducing personnel and facility costs. Anesthesia groups bill out their physicians by the minute, so our product delivers measurable cost savings to the hospital outside of neurosurgery. Simplifying and shortening a MIS procedure is very attractive to personnel involved with billing and finances, so **we consider this group to be our next most important customer group** after neurosurgeons.

Most hospitals have a Purchasing or Technology Review Board or similar to review requests for new devices and technology. These boards are typically diverse, but draw representatives from most of the groups listed above. Once a customer is willing to champion a device, they must convince a review board on factors such as cost savings, better patient outcomes, lower infection rates, and superior usage factors. At many institutions, a neurosurgeon is likely to get desired equipment fairly easily, whereas at others it can be a 3-6 month process. We intend to pursue initial customers at hospitals with favorable purchase policies.

C.2. Market.

C.2.1. Current Market. Numerous indications and diseases are currently treated using MIS techniques, including DBS placement (2-4), epilepsy and tremor ablation(26-28), guided biopsy(29,30), and cryoablation of prostate and abdominal vascular malformations(31-34). The number of neurosurgical procedures performed each year is relatively small, estimated at 1250-1600 patients/year. We infer this estimate from publicly available figures for ClearPoint revenue and procedure prices. At most, this corresponds to a market size for AccuGyd of $6.5M. However, we are optimistic that there is abundant room for growth in the number of procedures for both current and nascent procedures.



	Cancer	Parkinson's	Epilepsy	Stroke	Total
TAM: USA Prevalence	25,000	100,000	50,000	100,000	275,000
SAM: Valid Usage Cases	25,000	8,000	15,000	30,000	78,000
SAM: ImgGyd revenue	$100M	$64M	$60M	$120M	$312M

Figure 1. Prevalence in the USA for indications which could be aided by minimal invasive neurosurgery. Total Addressable Market (TAM) indicates overall prevalence. The Serviceable Available Market (SAM) indicates the subset of cases that will realistically benefit from treatment. The bottom row shows projected revenue from the SAM valid usage cases, assuming revenue of $4000 per procedure. Note that the TAM and SAM market sizes far outstrip current ability to provide these services.

C.2.2. Growing and Future Markets. MIS techniques support numerous upcoming clinical trials in Parkinson's disease(7, 35, 36), neurodegenerative diseases, and rare childhood "orphan" diseases. New intraparenchymal gene therapies in Phase II trials could be applicable to every case of PD at disease onset(36). The clinical REASSESS trial is underway to examine using MIS to remove clot in intracerebral hemorrhagic (ICH) stroke.

ICH stroke affects about 100,000 people per year in the USA, with about 30,000 patients per year being able to realistically benefit from a MIS-based clot evacuation(37). Prevalence is much higher in Japan. In terms of market size, these growing markets have the potential to greatly eclipse the current markets in C.2.1.

Gene therapy is poised to become a large market for image-guided neurosurgeries. Humans suffer from over 6,000 monogenic diseases, with devastating effect when unexpected neurodegeneration affects an otherwise healthy child or teen. Losses in gait, vision, speech, and missed developmental marks slowly unveil a diagnosis that profoundly alters a family for life. Rare neurological diseases such as Batten, Tay-Sachs, MPS III, Gaucher, and numerous ataxias frequently manifest just as parents expect to see their child's potential unfold, forever disrupting an entire family's future. Though these "orphan diseases" each affect fewer than 200,000 people nationwide, they are "collectively common," according to a recent *Lancet* editorial(38), affecting more than 300 million people globally. They are thus an important therapy target.

Rapid advances in genetic testing (see letter from Tom Miller, Bayer) are driving down the cost of genetic assays so it is feasible that it will be routine to test the entire infant genome within the next decade, enabling these rare diseases to be identified before they begin to damage the brain. The ability to detect rare neurodegenerative diseases is here already and will become increasingly available at reasonable price points.

Advances in molecular biology, gene therapy, and gene editing coupled with successful business and regulatory incentives have fueled a burgeoning pharmaceutical industry tailored to rare diseases that is economically successful and very popular within venture capital circles. In diseases treatable by "gain of function," where a gene only needs to be added instead of being edited, academic and pharmaceutical companies have demonstrated how established gene therapy approaches can restore metabolic function in focal treatments of the brain(39) or spinal cord. Significant improvements have been seen when the necessary genetic correction is within a localized brain region such as the substantia nigra(40). The biggest gaps in curing rare neurodegenerative diseases are not due to the small prevalence of any one disease.

Congressionally mandated FDA incentives for development of drugs to treat rare pediatric diseases(41) and orphan diseases(42) spur pharmaceutical companies in development in these areas. Rare Pediatric Disease Priority Review Vouchers may be sold or used by the sponsor for priority review of a different product. Orphan Drug Designation confers tax credits for clinical trials, user fee exemption, and a potential seven years of market exclusivity. **These regulatory advantages provide major incentives for development of therapies for orphan and rare pediatric diseases which would otherwise have too few patients to be financially viable.**

Currently many firms provide palliative treatments(43) that periodically administer enzymes intraventricularly or intrathecally to mitigate the brain damage caused by toxic metabolites that should be processed by the missing gene. However, a single, curative gene therapy would be a better solution(39). With the therapeutic solution already identified for some disorders, regulatory and technical roadblocks, not effectiveness, present the roadblock. Peter Marks, Director of the FDA's Center for Biologics Evaluation and Research (CBER), announced an **Operation Warp Speed-like pilot program for rare disease therapies** with enrollment starting in January 2024(44). At least 12 trials are underway to develop cures for rare neurodegenerative genetic diseases.

A key missing element is the technology to translate localized intraparenchymal adult therapies to whole brain pediatric treatments. **These cures need a guidance platform sized for the pediatric human brain** and catheters that can emit viral vectors over their length instead of just at their tip. ImgGyd LLC and its partners in porous catheters (OCCAM) and interstitial infusion modeling and monitoring (Therataxis) are positioning themselves to be leaders in this push for pediatric trials of new gene therapies. Dr. Walter Block and Dr. Raghu Raghavan (see Therataxis LOS) are inaugural members of the FDA-supported Critical Path Consortium on Lysosomal Storage Diseases, an umbrella that includes several neurodegenerative diseases. Dr. Thomas Miller, Global Pediatrics of Bayer (see LOS), advocated for strategies to overcome these hurdles.

These pharmaceutical efforts, especially once human trials are underway, are in general well-funded. Aside from the beneficial exposure, ImgGyd would derive significant revenue from participating in these trials. Successful trials would in turn create preferred and often required protocols for administration, securing ImgGyd a revenue stream. Strong patient and family support and advocacy organizations replace formal marketing and ensure enrollment of appropriate patients.

C.2.3. The Total Addressable Market (TAM) for gene therapy neurosurgery applications is difficult to assess due to the large number of orphan diseases with small populations and the potentially explosive growth if certain gene therapy hurdles can be cleared. It is more useful to consider the Serviceable Available Market (SAM) for several representative diseases. For example, Batten Disease patients number about 14,000 worldwide, with an age range of 2-16. This yields about 1000 new cases per year. The frequency of administration of genetic therapies varies widely, but the goal is a single curative administration, which we will assume for this market size estimation. For all of the therapies we are aware of, the preferred administration route is direct injection into intraparenchymal space, which typically requires (or at least greatly benefits from) accurate image guidance, such as from AccuGyd. There are 4 other orphan diseases (Sachs, MPS III, Gaucher, and various ataxias) with similar patient populations, yielding about 5,000 new cases per year in aggregate. The MIS procedures tend to be complex, with 6 or more access points. This equates to 30,000 device placements per year. For these complex surgeries with multiple placements, AccuGyd would be positioned as a premiere procedure solution, with a per-patient pricing structure on the order of $20,000, leading to a **SAM of $60 million/year for gene therapy**.

It is difficult to predict the future, but gene therapy applications will play a major role within the 5-10 year timeframe. Our likely timeframe to receive FDA 510(k) clearance 4 years from the start of this Phase II project aligns well with the onset of many relevant clinical trials. Even if our assumptions and estimates of market size are incorrect, the revenue and credibility generated from participating in clinical trials will create flexibility that will allow us to minimize rounds of external funding and place us in a much stronger position for subsequent platform development and commercial sales.

<u>C.2.4. Bottom-Up vs. Top-Down.</u> The top-down analysis in the previous section C.2.3 indicates a potentially large and lucrative marketplace. However, a large market does not automatically guarantee large sales for a specific product. Instead, we must start with a bottom-up approach: How will we make our first sale, the next 3 sales, and so on? We have the optimistic assumption that we can create a small base of enthusiastic customers and capture an increasingly larger share of the market based on our product excellence, optimism shared by every successful startup. Here we discuss **concrete plans for acquiring early customers for subsequent growth**.

<u>C.2.4.1. Pre-clinical Applications.</u> Our first image guided surgery research experience was with non-human primates. While the low volume of procedures cannot justify a profitable business plan, supporting researchers in this area provides valuable experience to improve our products for humans. The AccuGyd technology is readily applicable to non-human primates, and FDA clearance is not needed here. Several of our pre-clinical research colleagues are interested in improving their image guided procedures by using AccuGyd once it becomes a usable product. We anticipate supporting 2-5 sites with about one procedure per month, recuperating only our costs. This pre-clinical experience will lend confidence to later regulatory and sales efforts.

<u>C.2.4.2. FDA 510(k) clearance</u> is required prior to commercial sales. We do not plan to seek early usage, e.g. under an IND, since we anticipate a relatively straightforward FDA clearance process, and we do not believe that surgeons will take the risk of using a non-cleared device for brain surgery.

<u>C.2.4.3. Initial Customers.</u> Our UW-Madison neurosurgery collaborators Dr. Ahmed and Dr. Lake, have indicated that they would like to be early customers. They are familiar with ClearPoint and AccuGyd and they firmly believe that AccuGyd is superior. Hence, UW-Madison Hospital will be our first sale. Dr. Ahmed will be our main sales spokesperson. He personally knows most of the neurosurgeons in the upper Midwest who perform image guided MIS procedures. Several of them have written enthusiastic letters of support. **This group will be our next five customers.** We intend to add new customers in a deliberate sequence, emphasizing excellent customer support and excellent outcomes. A single bad outcome at this stage can spell disaster, so this stage cannot be rushed. Once we achieve a solid toe-hold, we will begin to advertise more aggressively through standard marketing routes. Dr. Ahmed will also step up his contacts with neurosurgeons nationwide, focusing on luminary sites and training sites for neurosurgeons.

<u>C.2.4.4. Procedures and Sales.</u> Based on our interviews with neurosurgeons with a variety of treated procedures and environments, we can reasonably estimate 2 procedures per month at each location for the first year. As confidence grows, 1 procedure per week is likely. A major benefit of MIS neurosurgeries is that it creates a “halo effect” for hospitals, who come to be regarded as regional experts, which in turn garners more patients from a larger catchment basin. As more patients are referred, the number of procedures is likely to increase. Furthermore, the demand for gene therapy procedures is likely grow well beyond current demand patterns. Busy neurosurgery centers with several neurosurgeons are likely to perform an image guided MIS procedure 5x/week.

Our major expense for early sites will be our Product Representative who will be on site for every procedure. Our goal is to eventually have AccuGyd be so easy to use and so reliable that onsite personnel are not needed. However, we know that early users will expect onsite guidance, which we will prioritize to forestall any negative experiences or perception of difficulty with the new device.

<u>C.3. Obstacles.</u> There are three main types of obstacles in developing any medical device: technical, regulatory, and financial. The technical obstacles are addressed in the Research Plan.

<u>C.3.1. Regulatory Obstacles.</u> Every medical device faces a major hurdle of being cleared for commercial sale by the FDA. This is regarded as a major advantage once clearance is obtained since it limits competition.

Successful management requires viewing regulatory issues as both an opportunity and a duty, with a clear plan for initiation, clearance, and ongoing adjustment and documentation throughout the product lifecycle. We intend to work closely with experienced regulatory affairs consultants such as NAMSA.

C.3.2. Financial Obstacles. Our next financial obstacle will be to develop funding for obtaining FDA 510(k) clearance. We will begin immediately to identify potential investors and start conversations about funding. We have received advice to de-risk our product as much as possible before seeking funding, so we will begin earnest discussions during the final year of this Phase II project. We identified a reasonable local network of potential investors during the sale of our previous company. We made formal investment pitches at several groups' headquarters, receiving good feedback and invitations to return with other products. **We feel that we are in a good position to seek funding from investors in the local Chicago-Madison-Minneapolis corridor, a strong ecosystem for health care technology and medical devices.**

C.4. Competition.

C.4.1. ClearPoint™ is the predominant device for MRI-guided neurosurgery. It is the most common device by far for minimal invasive neurosurgeries when accurate device placement requires MRI location and validation during the procedure. The mechanism and workflow are explained in detail in Section A.2 and in the Research Plan (Section 3). Here we discuss the business-related aspects of competition with ClearPoint.

ClearPoint has raised over $100 million in external funding during its private and publicly traded period over the past 12 years. Its market capitalization, now $165 million, reached as high as $450 million when partnerships with certain pharmaceutical providers in the rare neurodegenerative space caused excitement. ClearPoint's debt position, and its repeated public statements that its product is well positioned, does not leave it with the flexibility to further invest in solving its workflow complexity and extended procedural times. Rather, it has clearly told its investors that profitability is its overriding purpose at this time. In essence, they are locked into their current technology because of regulatory and financial constraints. Worse, **the ClearPoint device is poorly positioned for pediatric gene therapy**, which is likely to become one of the major uses for image guided neurosurgeries.

C.4.2. Other Guidance Approaches. Many stereotactic OR solutions use pre-operative MRI when therapeutic monitoring is not necessary or when electrodes are placed while awake. Devices in this space include StarFix™, the Medtronic Stealth™ system, VarioGuide (BrainLab's alignment system), the Rosa™ robot (Medtech), Vertec™(X-ray guidance for bone procedures), and Neuroblate™ (Monteris) for laser ablation(45). None of these solution-specific devices provide a platform for asleep guided device placement or therapeutic image monitoring. MRI provides excellent tissue contrast, location accuracy, and the ability to make adjustments on the fly. Importantly, the overall cost per hour of a standard MRI suite is similar to or less than the cost of an OR suite. Thus, the field is moving toward performing any procedure in the MRI suite if placement precision (e.g. asleep DBS placement) or therapeutic monitoring (epilepsy ablation and gene therapy infusion) are essential (46).

C.4.3. CT image guidance is emerging as a viable competitor to MRI-guided procedures, enabled by sophisticated software that registers prior MRI scans into the CT space, thus showing detail-rich MRI data overlaid onto current CT images. It is unlikely that either MRI or CT will completely displace the other modality for image guidance applications, since each provides unique information for specific applications. ImgGyd is thus pivoting from viewing AccuGyd as only an MRI-focused guidance device, to development for both MRI and CT. We recognize that many of the imaging and guidance principles are the same across these modalities, so we are expanding our development and validation efforts to include CT.

C.5. Our Competitive Advantage. The technology embodied in AccuGyd was designed from the beginning to address shortcomings and frustrations in the predominant market leader, ClearPoint, as pointed out by neurosurgeons who work with ClearPoint. Technologically and functionally, AccuGyd has the potential to be a far superior product and to deliver a far superior user experience. It successfully addresses each of the eight Gaps identified in Section A.3. To summarize AccuGyd's advantages laid out throughout this proposal: AccuGyd will significantly reduce procedure time, reduce procedure complexity, facilitate a dependable procedure time without multiple unpredictable alignment attempts, and provide a much smaller footprint that is more compatible with pediatric patients. These benefits come with an already demonstrated similar accuracy as ClearPoint. The reduced complexity of the AccuGyd design will carry over to a reduced manufacturing cost, which we can use to either reduce the price to the customer for a competitive advantage, or use for increased profit. AccuGyd was designed to fit into existing neurosurgical workflows, so existing customers of ClearPoint can see an easy path to switching guidance devices. To summarize: **better, faster, and cheaper is always a competitive advantage**.

C.5. Regulatory Strategy. We present an abridged version of the Regulatory Strategy which can be found in the Research Plan, section 4.5. Here, we focus on elements that affect our business development and marketing plans. Since we believe that we will be able to use the ClearPoint device as a valid predicate for AccuGyd, **we**

will develop a comprehensive plan for application for FDA 510(k) clearance, including a detailed step-by-step plan, best practice approaches for developing software and hardware, and estimates for effort, time, and funds required. We will work with NAMSA, a MedTech CRO firm with experience in regulatory affairs and quality lifecycle management. We are requesting TABA funding for this aspect of our business development.

Our NAMSA consultants concur that an FDA 510(k) approach is justifiable, and they will support us in using the 510(k) pathway to market. Our latest research and our NAMSA consultants indicate that we will not need to undertake a clinical trial prior to FDA clearance. Rather, we can rely on the approach taken by ClearPoint™, which relied largely on bench testing using gel phantoms of human heads. Gel phantoms have long been a mainstay of quantifiable validation of new neurosurgical image-guided products and procedures. Gel phantoms and human cadaver heads are used routinely in training neurosurgeons on new procedures.

As part of the regulatory strategy, NAMSA experts will advise on the types of nonclinical testing required, particularly on the specific tests and documentation for accuracy and precision. NAMSA will help us develop a formal testing protocol that aligns with FDA guidance and requirements (see Research Plan section 4.5).

D. Intellectual Property (IP) Protection. The Wisconsin Alumni Research Foundation (WARF), the IP arm of UW-Madison, holds a patent entitled "Systems and Method for Surgical Guide For Real-Time MRI Surgery" (U.S. patent 11,179,056), the subject invention of this proposal. Drs. Ahmed and Oakes are among the inventors. We intend to license this technology from WARF during and after Phase II development (see WARF LOS).

E. Finance Plan. Our finance plan is straightforward: 1) Obtain NIH Phase II funding. 2) Obtain bridge funding for business development from federal and state resources ($150k) including the Wisconsin State Economic Engagement and Development (SEED) grant program. 3) Seek investor funding for FDA clearance ($1M). 4) Seek investor funding to begin production and sales ($500k). 4) Ramp sales up to achieve profitability. When the first round of investor funding is required for FDA clearance, the AccuGyd technology will have been largely de-risked. Upon obtaining FDA clearance, the marketing and sales aspects will be de-risked. Each event that removes risk serves to greatly add to the value of the product and the company, which in turn increases investor enthusiasm. Past success in the Midwest medical imaging industry has already prompted interest from the VC community for funds and expertise to build sales channels and distribution. We have a methodical and disciplined approach to growing the company, bearing in mind that we have additional products under development. We have at least 4 luminary MIS sites who were advocates in Phase I and Phase II who would like to become customers after we receive FDA clearance (see LOS), so we can formulate a plan around these early sales.

F. Production and Marketing Plan

F.1 Production of AccuGyd. We intend to maintain the current approach to production of AccuGyd components via 3D printing. This is cost-effective and yields a product with the desired specifications. Our current manufacturer, Midwest Prototyping, has indicated a willingness to ramp up their production capabilities to accommodate a steady supply for large customers. As we scale up, we will investigate injection molding. We will develop the sterile packaging and hardware kits in Aim 5. As we begin commercial sales, we will acquire warehouse space and dedicated personnel to handle storage, packaging, and shipping. We regard the manufacturing process as very straightforward, with local advisors to help us work through each step.

F.2 Marketing Approach. As discussed in Section C.2.3.3 (Initial Customers), we intend to gain our first customers through personnel contacts of our neurosurgeon collaborators. We learned from experience with our previous company that rather than for us to be our own advocates and salespeople, it is far preferable to have our neurosurgeon collaborators and customers advocate for us through their personal connections. Once we are confident in our customer support and production capabilities, we will focus on luminary sites and neurosurgery training sites, again relying on surgeon-to-surgeon outreach efforts. The neurosurgery community is relatively small and close-knit so this is a realistic plan. We will also advertise through standard channels including trade publications and at scientific conferences. However, we believe that most of our new customers will be through direct contact of neurosurgeons.

G. Revenue Stream

Our revenue will be based on selling AccuGyd as a single-use disposable medical device. We plan to charge $4000 per procedure. Our cost of supplies per procedure will be about $200 for the AccuGyd plastic components and $200 for the supporting hardware kit. The cost of materials is almost negligible. A billing code and charge mechanism will be created for each new installation of an AccuGyd base and for each additional AccuGyd cassette. The laptop for the GUI display will keep track of the procedures and send out billing information as well as keep track of supplies that need to be replenished. Initial sites are likely to be moderately active, with about 1 procedure per week. **For 5 sites, revenue will be $1 million/year, more than enough to cover operating expenses and to fund ongoing development efforts.**